2025 IEEE International Conference on Energy Technologies for Future Grids
Wollongong, Australia, December 7-11, 2025

Physics-Informed Neural Power Flow Solver for Three-Phase Unbalanced Distribution Networks

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2025 IEEE International Conference on Energy Technologies for Future Grids (ETFG) | 979-8-3315-7640-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ETFG61999.2025.11400950

***Abstract*—Power flow analysis is fundamental for the reliable operation and planning of modern distribution networks (DNs). However, conventional iterative numerical solvers are often computationally intensive, especially for large networks. In this work, we propose a Physics-Informed Graph Neural Network (PIGNN-3Φ) approach to speed-up the power flow calculations in three-phase unbalanced DNs. The proposed method directly exploits the network topology and learns the mapping from power injection to bus voltages while respecting the underlying physical laws. Physical consistency is enforced by embedding Kirchhoff's laws and Ohm's law relations into the model objective function, promoting only feasible and interpretable solutions while reducing dependence on large, high-quality labeled datasets typical of black-box deep learning approaches. Furthermore, we quantify model predictive uncertainty using the Monte Carlo dropout technique, which approximates Bayesian inference in neural networks and provides well-calibrated estimates of the model's epistemic uncertainty. The results show that the proposed PIGNN-3Φ outperforms black-box models and achieves an approximately $50\times$ computational gain over traditional numerical PF solvers while achieving comparable accuracy. Furthermore, the framework enables risk-aware decision making by providing calibrated predictive intervals for voltage estimates, enabling system operators to assess probabilities of constraint violations.**

***Index Terms*—Power flow analysis, physics-informed machine learning, distribution networks, uncertainty quantification.**

I. Introduction

The primary objective of power flow (PF) analysis in distribution networks (DNs) is to determine the bus voltages and line flows under specified loading conditions. Distribution system operators rely on these calculations to identify potential bottlenecks within their network and make informed decisions regarding grid planning and network upgrades. However, conventional numerical PF solvers, such as the Newton–Raphson method [1], are computationally intensive, particularly for performing large-scale planning studies, such as hosting capacity analysis, which often require millions of iterative PF calculations. Furthermore, even small modifications in grid topology necessitate a full recalculation of the numerical PF solution, further compounding the computational challenges.

Deep learning (DL)-based approaches offer a promising alternative to accelerate PF calculations by learning non-linear patterns from historical data [2], [3]. However, their black-box nature and reliance on extensive training datasets often limit their ability to model the intricate dynamics of electrical networks. Graph Neural Networks (GNNs) have emerged as a powerful tool for PF analysis, leveraging the inherent graph like structure of power grids to model diverse topologies [4]. Through message-passing, GNNs enable buses to exchange information with their neighbors, capturing complex electrical interactions within the network. For instance, the study in [5] employs a Graph Attention Network (GAT) to enhance graph-based learning by assigning weights to nodes through an attention mechanism, which effectively prioritizes critical nodes. However, most of the existing DL approaches are physics-agnostic, and their accuracy heavily depends on the quantity and quality of the training data. This underscores the need for hybrid approaches that integrate domain knowledge into the neural networks to ensure robustness and accuracy.

Physics-Informed Neural Networks (PINNs) are gaining increased attention in the scientific research community as hybrid approaches that combine data-based learning with underlying physical principles. Recent works, such as in [6], highlight the potential of PINNs to accelerate PF computations with improved generalization performance. Additionally, PINNs have also been utilized for other power system-related tasks such as line parameter estimation, state estimation, and fault detection [7], [8]. The study in [9] incorporates Kirchhoff's laws and system topology as auxiliary constraints to enhance PF accuracy and robustness. Similarly, the authors in [10] proposed a specialized GNN architecture, termed PowerFlowNet, which demonstrated comparable performance to the traditional Newton-Raphson method while significantly reducing the computational speed. However, most of the existing PINN-based methods for PF analysis are tailored for transmission networks, which differ significantly from DNs characterized by their radial topology, high R/X ratios, and unbalanced phase loading. Some approaches oversimplify the problem by focusing only on single-phase or balanced three-phase networks, thereby neglecting the complexities of modeling phase unbalance inherent in real-world networks. Therefore, there is a pressing need for better physics-aware methodologies that can accurately perform PF calculations in unbalanced DNs while maintaining computational efficiency.

Uncertainty quantification (UQ) is vital for PF analysis as it quantifies confidence in model predictions and enables risk-aware decision-making under the high penetration of distributed energy resources and fluctuating demand. Monte

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Carlo (MC) dropout [11] provides a scalable approach for obtaining probabilistic outputs from neural networks, offering a practical alternative to computationally intensive Bayesian inference. The study in [12] employs MC dropout within a GNN model to enable probabilistic voltage predictions with quantified uncertainty. Despite their potential, the application of probabilistic PINNs in power systems studies remains relatively underexplored and requires further investigation.

Building upon our prior work in [6], this paper extends the application of PINNs for solving the probabilistic PF problem in three-phase unbalanced DNs. We present a novel physics-informed graph neural network (PIGNN-3Φ) framework that combines the computational efficiency of data-driven methods with the physical principles based on PF equations. This ensures accurate and physically consistent PF solutions. The key contributions of this work are:

- A probabilistic PIGNN-3Φ model for performing the accurate PF calculations in three-phase unbalanced DNs.
- A physics-informed loss function incorporating Kirchhoff's law (KCL) and Ohm's law as soft constraints to ensure physical consistency and feasible solutions.
- MC dropout-enabled probabilistic inference for quantifying uncertainty in model predictions.

The remainder of this paper is organized as follows: Section II provides an overview of the PF formulation for radial DNs. In Section III, we discuss the proposed methodology in detail. Experimental results are provided in Section IV, followed by the conclusion in Section V.

II. Power Flow Formulation

The DistFlow model, as described in [13], provides an accurate representation of power flows in radial DNs. In this section, we discuss the DistFlow formulation and its linearized form for three-phase unbalanced systems (i.e., LinDist3Flow). Let $\mathcal{N} = \{0, 1, \ldots, n\}$ denote the set of buses, and $\mathcal{E} \subseteq \mathcal{N} \times \mathcal{N}$ be the set of electrical lines in the simplified radial DN, as shown in Fig. 1. For each bus $n \in \mathcal{N}$, the squared voltage is denoted as $v_n = |V_n|^2$. For each line $(\pi_n, n) \in \mathcal{E}$, the line impedance is $z_n = r_n + jx_n$, with resistance r_n and reactance x_n. The complex power injection at bus n is $s_n = p_n + jq_n$, and the squared current magnitude is $\ell_n = |I_n|^2$.

The non-linear DistFlow equations are expressed as:

$$v_n = |V_n|^2, \qquad \ell_n = |I_n|^2 \tag{1}$$

$$\sum\nolimits_{k:n\to k} P_k = p_n + P_n - r_n \ell_n \tag{2}$$

$$\sum\nolimits_{k:n\to k} Q_k = q_n + Q_n - x_n \ell_n \tag{3}$$

$$v_n = v_{\pi_n} - 2(r_n P_n + x_n Q_n) + (r_n^2 + x_n^2)\ell_n \tag{4}$$

$$v_{\pi_n}\, \ell_n = P_n^2 + Q_n^2 \tag{5}$$

The equations (2)-(5) represent the nodal power balance and voltage drop relations in a single-phase radial network. This formulation is non-linear due to the quadratic relation in (5). By neglecting the loss terms $r_n\ell_n$, $x_n\ell_n$, and $(r_n^2 + x_n^2)\ell_n$, the model reduces to its linearized form (i.e., LinDistFlow). This linearized formulation is widely used in planning and control studies as it offers improved analytical tractability while introducing only minor approximation errors. For three-phase unbalanced systems, these equations can be expressed in a compact matrix notation, where phase-wise voltage and line power flow variables are defined as:

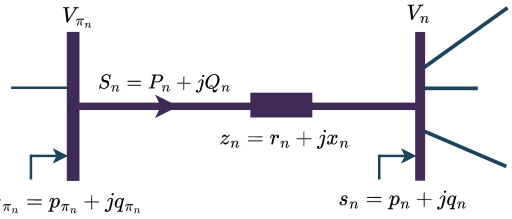


Fig. 1: A simple two-bus distribution system where an electrical line with impedance z_n connects bus n to bus π_n.

$$\mathbf{V} \in \mathbb{R}^{3|\mathcal{N}|}, \quad \mathbf{P}, \mathbf{Q} \in \mathbb{R}^{3|\mathcal{E}|}.$$

For each branch $(i, j) \in \mathcal{E}$, we introduce the parameter $\kappa = \sqrt{3}$ and define the per-branch resistance and reactance coupling matrices as:

$$\mathbf{M}_R^{(ij)} = \begin{bmatrix} R_{aa} & R_{ab} - \kappa X_{ab} & R_{ac} + \kappa X_{ac} \\ R_{ba} + \kappa X_{ba} & R_{bb} & R_{bc} - \kappa X_{bc} \\ R_{ca} - \kappa X_{ca} & R_{cb} + \kappa X_{cb} & R_{cc} \end{bmatrix},$$

$$\mathbf{M}_X^{(ij)} = \begin{bmatrix} X_{aa} & X_{ab} + \kappa R_{ab} & X_{ac} - \kappa R_{ac} \\ X_{ba} - \kappa R_{ba} & X_{bb} & X_{bc} + \kappa R_{bc} \\ X_{ca} + \kappa R_{ca} & X_{cb} - \kappa R_{cb} & X_{cc} \end{bmatrix}.$$

The per-branch coupling matrices are then assembled into the following block-diagonal matrices:

$$\mathbf{M}_R := \mathrm{blkdiag}\{\mathbf{M}_R^{(ij)}\}, \quad \mathbf{M}_X := \mathrm{blkdiag}\{\mathbf{M}_X^{(ij)}\}.$$

Equivalently, the LinDist3Flow voltage drop (Ohm's law) and power balance equations can be written in compact matrix form using the branch–bus incidence matrix $\mathbf{A}$ as:

$$(\mathbf{A} \otimes \mathbf{I}_3)\mathbf{V} = 2\mathbf{M}_R\mathbf{P} + 2\mathbf{M}_X\mathbf{Q}, \tag{6}$$

$$(\mathbf{A}^\top \otimes \mathbf{I}_3)\mathbf{P} = \mathbf{p}_{\mathrm{net}}, \quad (\mathbf{A}^\top \otimes \mathbf{I}_3)\mathbf{Q} = \mathbf{q}_{\mathrm{net}}, \tag{7}$$

where $\mathbf{p}_{\mathrm{net}}, \mathbf{q}_{\mathrm{net}} \in \mathbb{R}^{3|\mathcal{N}|}$ denote the per-phase net injections. The Kronecker product $\mathbf{A} \otimes \mathbf{I}_3$ extends the branch–bus incidence matrix $\mathbf{A} \in \mathbb{R}^{|\mathcal{E}|\times|\mathcal{N}|}$ to three phases, effectively modeling mutual phase coupling within each line segment. This formulation provides a computationally efficient and convex representation that is analytically tractable for large radial networks and sufficiently accurate under moderate voltage drops and line losses. To ensure physical consistency in the proposed PIGNN-3Φ model's predictions, we embed the LinDist3Flow relations (6)–(7) into the training objective as soft constraints. This helps the model learn a physically plausible PF solution by penalizing violations of Ohm's law and the power balance equations. This physics-guided regularization improves model generalization to out-of-distribution operating conditions and increases its robustness to measurement noise during training.

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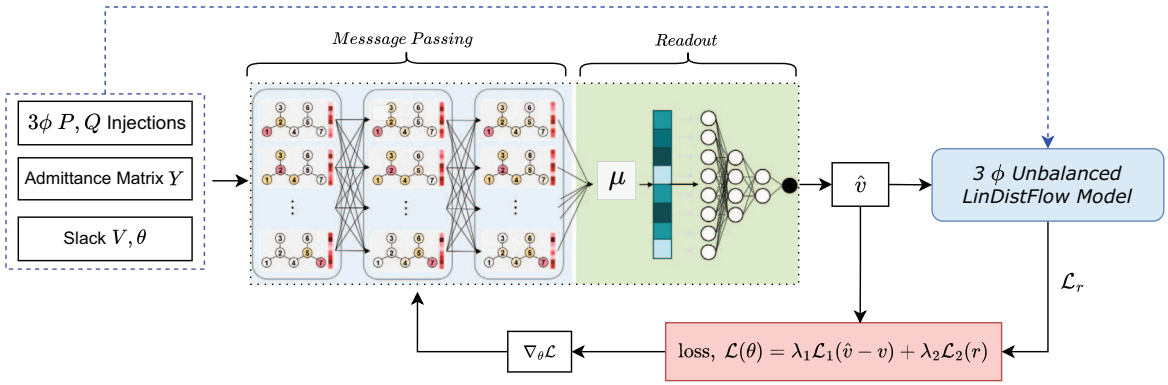


Fig. 2: Proposed PIGNN-3Φ model architecture. It processes three-phase input power injections and admittance matrix through GAT layers, leveraging message passing to learn complex node interactions. The readout stage outputs predicted voltages $\hat{v}$. To enforce the physical consistency, residuals $\mathcal{L}_r$ are added to the overall loss $\mathcal{L}(\theta)$, combining data-fit and physics constraints.

III. Proposed Methodology

In this study, we propose the PIGNN-3Φ model to predict probabilistic voltages in three-phase unbalanced DNs. The primary aim is to produce fast, accurate approximations of the three-phase unbalanced PF problem while providing UQ. By explicitly modeling phase unbalance and embedding the physical constraints into the learning objective, the framework achieves improved generalization to unseen operational conditions and substantially reduces computational cost compared to iterative solvers. Fig. 2 shows the overall architecture of the proposed PIGNN-3Φ model, which comprises two key components: (i) a graph-based architecture, and (ii) a physics-informed loss function that employs the LinDist3Flow equations. The proposed model takes per-phase power injections, the admittance matrix, and slack bus conditions as inputs to predict voltages at all non-slack buses. The architecture incorporates a multi layer GAT [5] mechanism to effectively model the influence among neighboring nodes during message passing, thereby capturing fine-grained phase-level interactions. The node embeddings, which represent the latent feature vectors, are aggregated via a readout module composed of Multi-Layer Perceptrons (MLPs)[1]. The readout maps the aggregated embeddings to the predicted per-phase voltage magnitudes $\hat{v}$. The model predictions are constrained by a physics-informed loss function, which seamlessly integrates physics-based residuals with the data-fit term to produce feasible solutions that adhere to the underlying physical principles.

A. Graph Attention Network (GAT)

We model the electrical distribution grid as an undirected graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, where $\mathcal{N}$ represents the set of nodes (i.e., physical buses) and $\mathcal{E}$ denotes the set of edges (i.e., power lines). Each node $i \in \mathcal{N}$ is associated with a feature vector $\mathbf{x}_i \in \mathbb{R}^d$, encoding electrical quantities such as target voltage magnitude $|V_i|$ and input power injections (p_i, q_i). Each edge $(i, j) \in \mathcal{E}$ carries a feature vector $\mathbf{e}_{ij} \in \mathbb{R}^k$, typically including line parameters such as resistance R_{ij} and reactance X_{ij}. The underlying mechanism of graph-based learning is the *message-passing* process, in which nodes iteratively exchange information with their neighbors to update their embeddings via three steps: (i) message computation, (ii) aggregation, and (iii) node update. GAT employs attention coefficients to dynamically assign importance weights to neighboring nodes during the message aggregation process. This mechanism is particularly advantageous in electrical networks, where the influence of one bus on another can vary significantly depending on the network topology and loading conditions. In each GAT layer l, messages $\mathbf{m}_{ij}^{(l)}$ are passed from node j to node i to captures both node-level and edge-level dependencies. The raw (unnormalized) attention scores $e_{ij}^{(l)}$ are computed by a shared attention mechanism and normalized across node i's neighborhood using a softmax function to obtain attention weights $\alpha_{ij}^{(l)}$. These weights rescale the messages based on their importance, which are then aggregated into $\mathbf{s}_i^{(l)}$. Finally, a learnable update function $U^{(l)}(\cdot)$, typically an MLP, produces the new embedding $\mathbf{h}_i^{(l+1)}$. The mathematical formulations of these message passing steps for a layer l are defined as:

$$\mathbf{m}_{ij}^{(l)} = \psi^{(l)}\left(\mathbf{h}_j^{(l)}, e_{ij}\right), \tag{8}$$

$$e_{ij}^{(l)} = \mathrm{LeakyReLU}\Big(\mathbf{a}^{(l)\top}\big[\,\mathbf{W}^{(l)}\mathbf{h}_i^{(l)} \,\|\, \mathbf{W}^{(l)}\mathbf{h}_j^{(l)}\,\big]\Big), \tag{9}$$

$$\alpha_{ij}^{(l)} = \frac{\exp\big(e_{ij}^{(l)}\big)}{\sum_{k\in\mathcal{N}_i}\exp\big(e_{ik}^{(l)}\big)}, \tag{10}$$

$$\mathbf{s}_i^{(l)} = \sum_{j\in\mathcal{N}_i} \alpha_{ij}^{(l)}\,\mathbf{m}_{ij}^{(l)}, \tag{11}$$

$$\mathbf{h}_i^{(l+1)} = U^{(l)}\left(\mathbf{h}_i^{(l)}, \mathbf{s}_i^{(l)}\right), \tag{12}$$

where $\mathbf{W}^{(l)} \in \mathbb{R}^{d'\times d}$ is a learnable weight matrix mapping applied to node features and $\mathbf{a}^{(l)} \in \mathbb{R}^{2d'}$ is the trainable attention vector. To enhance the model's expressiveness, we employ a *multi-head attention* strategy. Instead of relying on

[1]An MLP is a feedforward neural network consisting of multiple layers.

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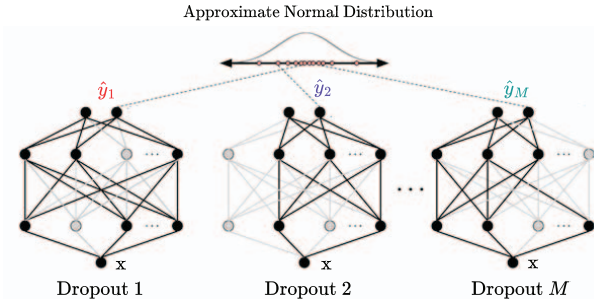


Fig. 3: Monte Carlo dropout process for uncertainty quantification. For a single input x, we take multiple stochastic predictions $\hat{y}$ to estimate the predictive mean and variance.

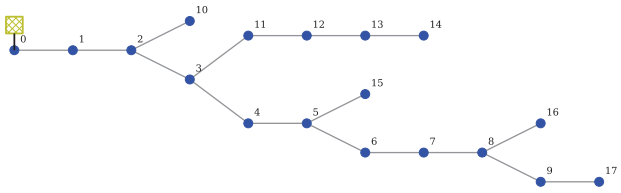


Fig. 4: Modified 18-bus CIGRE LV test network.

a single attention mechanism, multiple attention heads are applied in parallel, with each head capturing distinct and intricate relational patterns among neighboring nodes.

B. Physics-Informed Loss Function

A key component of the proposed model is its physics-informed loss function, which combines a data-fit term with a physics-based loss term derived from the LinDist3Flow equations. The constraint residuals are defined as:

$$\mathbf{r}_{\text{Ohm}} = \mathcal{A}\,\hat{\mathbf{v}} \;-\; 2\mathbf{M}_R\,\hat{\mathbf{P}} \;-\; 2\mathbf{M}_X\,\hat{\mathbf{Q}}, \tag{13}$$

$$\mathbf{r}^P_{\text{KCL}} = \mathcal{A}^\top\hat{\mathbf{P}} - \mathbf{p}_{\text{net}}, \qquad \mathbf{r}^Q_{\text{KCL}} = \mathcal{A}^\top\hat{\mathbf{Q}} - \mathbf{q}_{\text{net}}, \tag{14}$$

where $\mathcal{A} = \mathbf{A} \otimes \mathbf{I}_3$. The above constraint residuals are then combined to form a unified physics-based loss term:

$$\mathcal{L}_r = \frac{1}{3|\mathcal{E}|}\left\|\mathbf{r}_{\text{Ohm}}\right\|_2^2 + \frac{1}{3|\mathcal{N}|}\left(\left\|\mathbf{r}^P_{\text{KCL}}\right\|_2^2 + \left\|\mathbf{r}^Q_{\text{KCL}}\right\|_2^2\right), \tag{15}$$

The final objective function is the weighted sum of data-fit loss and physics residual terms:

$$\mathcal{L}(\theta) \;=\; \lambda_1\,\mathcal{L}_1(\hat{v} - v_{true}) \;+\; \lambda_2\,\mathcal{L}_{\text{r}}, \tag{16}$$

where $\lambda_1, \lambda_2 > 0$ are tunable hyperparameters that balance the contributions of the data-fit and the physics loss terms. This formulation ensures that the model remains faithful to the observed measurements while enforcing the three-phase PF constraints. Specifically, Ohm's law and the power balance constraints serve as a soft regularizer that restricts the model's feasible solution manifold and steers it toward physically admissible regimes. This enforces a principled trade-off between empirical fit and physical consistency, accelerating convergence and improving generalization. Moreover, the physics-based loss terms enable the network to capture cross-phase dependencies through the matrices $\mathbf{M}_R$ and $\mathbf{M}_X$.

C. Uncertainty Quantification Using MC Dropout

Bayesian inference provides a principled framework for modeling predictive uncertainty, which is essential for risk-aware operation in DNs. Quantifying uncertainty in predicted voltages enables operators to implement chance-constrained control and to assess the risk of voltage limit violations. The MC dropout approach, as demonstrated in [11], approximates Bayesian inference in neural networks by interpreting dropout as a variational approximation to a deep Gaussian process. While dropout is traditionally used during model training to prevent overfitting by randomly deactivating a subset of hidden units, MC dropout extends this mechanism to inference. During inference, we approximate the posterior predictive distribution by performing M stochastic forward passes through the model, each employing an independently sampled dropout mask drawn from a Bernoulli distribution. This procedure yields an ensemble of voltage predictions for each input instance. The empirical predictive mean and variance are then computed to approximate the predictive distribution and to quantify model confidence. Fig. 3 illustrates this process, where M dropout masks are applied to output the predictive distribution of the target y for the same input x.

IV. Experimental Setup

We evaluated the proposed PIGNN-3Φ model on the 18-bus Modified CIGRE LV unbalanced distribution feeder, as detailed in [14]. The feeder layout is shown in Fig. 4. To generate the training dataset, we adopted an MC sampling strategy commonly used in PF studies. Specifically, active and reactive power profiles for residential customers were first sampled from a pool of standard load curves. To introduce realistic variability in the load patterns, the sampled profiles were perturbed with additive Gaussian noise (standard deviation 15%), applied independently across phases and time steps. Numerical PF simulations were subsequently performed using the `pandapower` framework [15] on the perturbed load profiles to obtain the corresponding voltage labels. We generated 2400 grid snapshots and split them into training, validation, and test sets using a $60/20/20$ ratio. The model was implemented in Python using the PyTorch and PyTorch Geometric libraries [16]. For benchmarking, we implemented two black-box neural network models: a GNN and a fully connected neural network (FCNN). Both baselines consist of 5 hidden layers with 128 units per layer and ReLU activations.

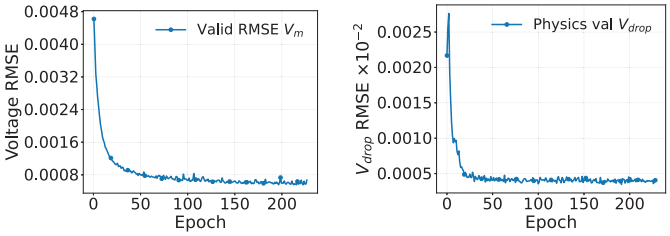
V. Numerical Results

The proposed PIGNN-3Φ model is compared against black-box GNN and FCNN models, using the Newton–Raphson PF solution as the baseline. The comparison evaluates four key aspects: (i) voltage prediction accuracy; (ii) calibration and quality of the MC dropout uncertainty estimates; (iii)

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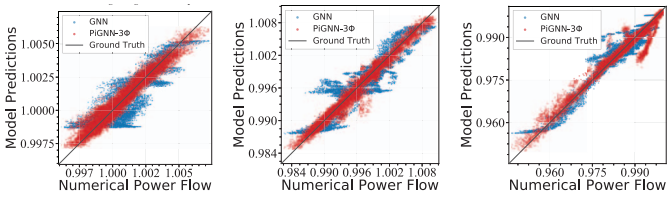
TABLE I: Performance comparison of different models. Lower RMSE and MAE, and higher R^2 indicate better performance.

Comparison Model	RMSE ($\times 10^{-3}$)				MAE ($\times 10^{-3}$)				R^2			
	Phase A	Phase B	Phase C	Overall	Phase A	Phase B	Phase C	Overall	Phase A	Phase B	Phase C	Overall
FCNN	1.100	2.200	2.600	2.000	0.900	1.600	2.000	1.500	0.780	0.840	0.930	0.850
GNN	0.830	1.910	2.310	1.680	0.660	1.410	1.710	1.260	0.805	0.856	0.942	0.867
PIGNN-3Φ	**0.450**	**0.930**	**2.170**	**1.180**	**0.340**	**0.620**	**1.160**	**0.710**	**0.945**	**0.966**	**0.949**	**0.953**



(a) Voltage RMSE (b) V_{drop} RMSE

Fig. 5: PIGNN-3Φ model validation set RMSE trends for voltages and V_{drop} (i.e., Ohm's law) during training.



(a) Phase A (b) Phase B (c) Phase C

Fig. 6: Comparing predicted voltages from both the GNN and PIGNN-3Φ models with ground truth across three phases.

robustness and generalization to previously unseen loading conditions; and (iv) inference time relative to iterative solvers. The model's accuracy is evaluated using several metrics, such as the coefficient of determination (R^2), the root mean square error (RMSE), and the mean absolute error (MAE). The R^2 metric quantifies how well the model's predictions align with true values by measuring the proportion of variance in the target explained by the model. The higher R^2 values indicate better model performance. Fig. 5 shows the decreasing validation RMSE trends over training epochs for both bus voltage magnitude and voltage drop (i.e., Ohm's law). These loss plots indicate that the PIGNN-3Φ model simultaneously learns data-fit and physics constraints, which results in stable training and fast convergence. In the following subsections, a detailed analysis of the experimental results is presented.

A. Voltage Prediction Accuracy

As shown in Table I, the PIGNN-3Φ model consistently achieves superior accuracy across all three evaluation metrics compared to the data-driven GNN and FCNN models. Fig. 6 presents scatter plots comparing the predicted and actual voltage magnitudes from the proposed PIGNN-3Φ and GNN for all three phases. The closer the data points are to the diagonal line in these plots, the better the predictive accuracy. The PIGNN-3Φ demonstrates better predictive performance

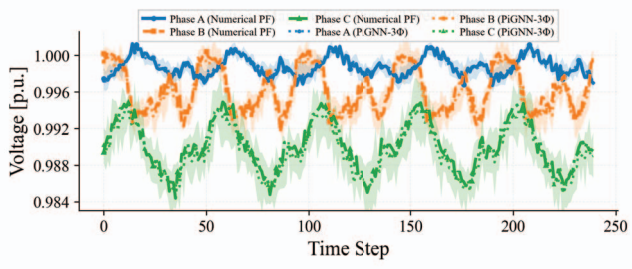
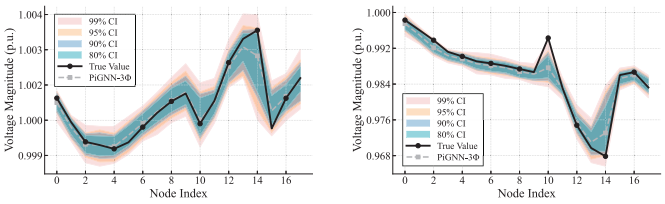


Fig. 7: Voltage predictions at Bus 3: Phase A (blue), Phase B (orange), and Phase C (green), with 95% confidence intervals (shaded regions). Solid lines indicate true values from the numerical solver; dashed lines represent PIGNN-3Φ predictions.



(a) Phase A (b) Phase C

Fig. 8: Calibrated predictive intervals generated by PIGNN-3Φ for voltage magnitudes across phases A and C of all nodes.

across three phases compared to the black-box GNN. This demonstrates the model's improved ability to capture complex phase-wise PF relationships and to learn phase unbalance more effectively. Moreover, the reduced dispersion in the predictions is largely due to the physics-informed constraints, which help the model reject physically implausible solutions and ensure consistency with PF principles. A sample probabilistic voltage prediction at bus 3 for each phase is shown in Fig. 7.

B. Calibrated Uncertainty under Chance Constraints

The calibration process in UQ is crucial because it ensures reliable interpretation of the model's predictive uncertainty. Calibration is performed on the validation set by comparing empirical coverage rates with nominal confidence levels. Using MC dropout, we compute multiple CIs for voltage predictions and calibrate them to align with the observed errors. Fig. 8 shows the calibrated intervals for all nodes. Narrower intervals (e.g., 80%) capture most true voltage values, indicating high predictive accuracy of the proposed model, while wider intervals (e.g., 99%) provide more conservative and robust coverage. Besides that, we also observed that the predictive intervals are systematically wider for nodes with very high loading or abrupt load variations (e.g., buses 12–14), indi-

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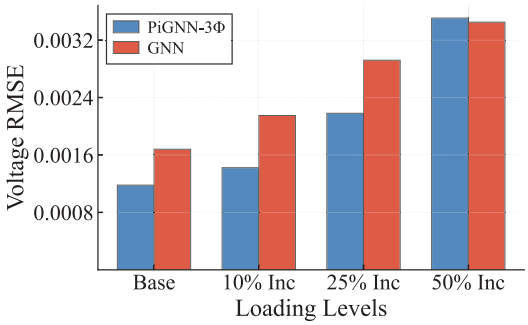


Fig. 9: Comparison between proposed PIGNN-3Φ and GNN model under different network loading conditions.

cating higher model uncertainty. By selecting an appropriate confidence level, operators can enforce probabilistic voltage limits that bound the probability of violations to a prescribed risk tolerance, thereby supporting risk-aware decision making.

C. Impact of Loading Variations

To assess the model's robustness under varying network loading conditions, we increased the active and reactive power injections at each bus by 10%–50% above their nominal values. The model, trained only on nominal load levels, was evaluated on unseen high-loading scenarios to further assess its generalization ability. As illustrated in Fig. 9, the RMSE of the proposed PIGNN-3Φ model increases less than that of the black-box GNN as loading levels increase. For minor load changes (+10%), the PIGNN-3Φ maintains high accuracy with minimal RMSE increase. However, for significant load changes (+25% and +50%), the RMSE rises more noticeably, reflecting the challenges of generalizing to extreme conditions. The proposed model handles moderate load variations well, but is more sensitive to heavy load scenarios; therefore, retraining might be needed for very high loading conditions.

D. Time Complexity

The PIGNN-3Φ model offers a significantly lower computational cost than traditional numerical PF solvers, making it well-suited for performing large-scale network expansion and planning studies. We compared the proposed model inference time with the numerical solver on two differently sized networks: the CIGRE 18-bus and the IEEE LV European 906-bus network. The numerical solver scaled poorly with increasing network size (i.e., 56.7 ms per sample for the 18-bus and 561 ms for the 906-bus). In contrast, PIGNN-3Φ achieves much better computational efficiency, with per sample inference time increased only marginally with network size (i.e., 4.5 ms for the 18-bus and 11.4 ms for the 906-bus), which provides up to a 50× speed-up. This experiment was performed on a machine with an Intel Core i7 CPU (16 cores) and 32 GB of RAM.

VI. Conclusion

We propose a physics-informed graph neural network for performing fast probabilistic power flow calculations in three-phase unbalanced distribution networks. By embedding constraints based on DistFlow equations into the objective function, the model ensures physically consistent predictions and effectively captures phase unbalance. We incorporated the MC dropout approach to enable probabilistic inference with uncertainty quantification, which provides reliable confidence intervals for risk-aware decision making. Experimental results show that the proposed model achieves accuracy comparable to traditional Newton-Raphson. Future work will focus on evaluating this approach on larger networks under partial observability, along with topological variations.

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